

Functions

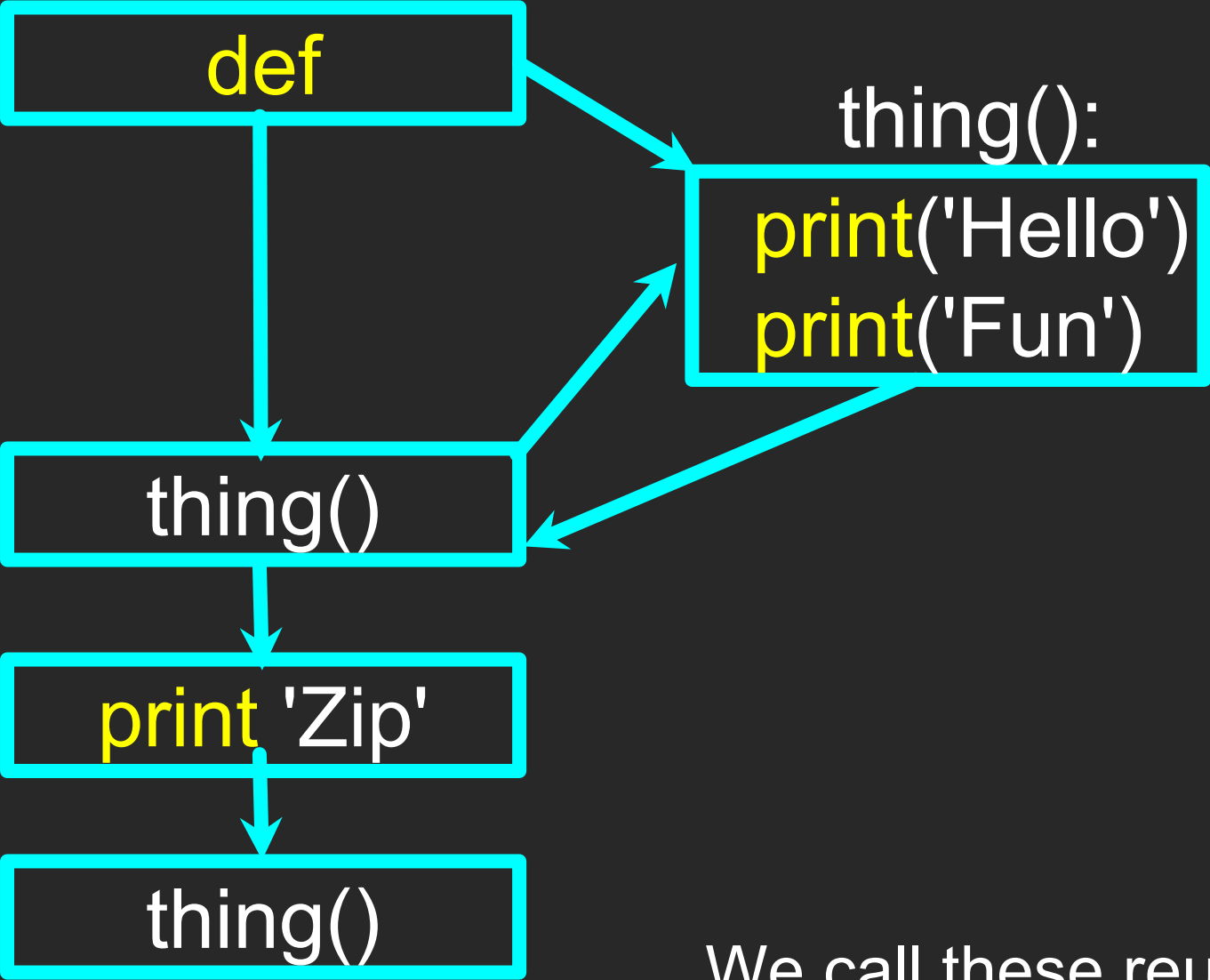
Chapter 4



Python for Everybody
www.py4e.com



Stored (and reused) Steps



Program	Output
<pre>def thing(): print('Hello') print('Fun')</pre>	
<pre>thing()</pre>	Hello Fun
<pre>print('Zip')</pre>	Zip
<pre>thing()</pre>	Hello Fun

We call these reusable pieces of code “functions”

Python Functions

- There are two kinds of functions in Python.
 - **Built-in functions** that are provided as part of Python - `print()`, `input()`, `type()`, `float()`, `int()` ...
 - **Functions that we define ourselves** and then use
- We treat the built-in function names as “new” **reserved words** (i.e., we avoid them as variable names)

Function Definition

- In Python a **function** is some reusable code that takes **arguments(s)** as input, does some computation, and then returns a result or results
- We define a **function** using the **def** reserved word
- We call/invoke the **function** by using the function name, parentheses, and **arguments** in an expression

Argument

Assignment

'w'

Result

`big = max('Hello world')`

```
>>> big = max('Hello world')
>>> print(big)
w
>>> tiny = min('Hello world')
>>> print(tiny)

>>>
```

Max Function

```
>>> big = max('Hello world')
>>> print(big)
w
```

'Hello world'
(a string)



max()
function



'w'
(a string)

A function is some stored code
that we use.

A function takes some input and
produces an output.

Guido wrote this code

Max Function

```
>>> big = max('Hello world')
>>> print(big)
w
```

'Hello world'
(a string)



```
def max(inp):
    blah
    blah
    for x in inp:
        blah
        blah
```



'w'
(a string)

A function is some stored code that we use.

A function takes some input and produces an output.

Guido wrote this code

Type Conversions

- When you put an integer and floating point in an expression, the integer is **implicitly** converted to a float
- You can control this with the built-in functions `int()` and `float()`

```
>>> print float(99) / 100
0.99
>>> i = 42
>>> type(i)
<class 'int'>
>>> f = float(i)
>>> print(f)
42.0
>>> type(f)
<class 'float'>
>>> print(1 + 2 * float(3) / 4 - 5)
-2.5
>>>
```


String Conversions

- You can also use `int()` and `float()` to convert between strings and integers
- You will get an **error** if the string does not contain numeric characters

```
>>> sval = '123'
>>> type(sval)
<class 'str'>
>>> print(sval + 1)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: cannot concatenate 'str'
and 'int'
>>> ival = int(sval)
>>> type(ival)
<class 'int'>
>>> print(ival + 1)
124
>>> nsv = 'hello bob'
>>> niv = int(nsv)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: invalid literal for int()
```

A Function of Our Own



Acknowledgements / Contributions



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